

DEFINITION — HOW DATA IS DESCRIBED

CONTRACTS6



ODCS

BITOL @ LF



OpenAPI

LF



AsyncAPI

LF



GraphQL

LF



gRPC

CNCF



OData

OASIS

DATA PRODUCTS4



ODPS

BITOL @ LF



DPDS

ODM



ODPSpec

LF



DPROD

OMG

SCHEMA5



XML Schema

W3C



JSON Schema

IETF



SQL DDL

ISO/IEC



AVRO Schema

ASF



Protobuf

GOOGLE

SEMANTICS6



OSI

OSI INITIATIVE



RDF/OWL

W3C



DCAT

W3C



SKOS

W3C



SHACL

W3C



JSON-LD

W3C

STORAGE — WHERE DATA LIVES

FILE FORMATS7



CSV

IETF



JSON

IETF



XML

W3C



PARQUET

ASF



AVRO

ASF



ORC

ASF



Lance

LANCE

OPEN TABLE FORMATS4



Iceberg

ASF



Delta

LF



Hudi

ASF



Lance

LANCE

STORAGE SYSTEMS2



S3

AWS



HDFS

ASF

MOVEMENT — HOW DATA FLOWS BETWEEN SYSTEMS

DATABASE CONNECTIVITY3



JDBC

JCP



ODBC

ISO/IEC



ADBC

ASF

INTERCONNECTION3



HTTP

IETF



FTP / SFTP

IETF



Delta Sharing

LF

MESSAGING4



Kafka

ASF



MQTT

OASIS



CloudEvents

CNCF



AMQP

OASIS

TRANSFORMATION — HOW DATA IS PROCESSED AND RESHAPED

IN-MEMORY FORMAT2



DataFrame API

DATA-APIS



Apache Arrow

ASF

PROCESSING6



dbt

DBT LABS



Spark

ASF



Beam

ASF



Pandas

NUMFOCUS



Ibis

IBIS



SQL DML

ISO/IEC

DISCOVERY — HOW DATA IS FOUND AND TRACED

CATALOG APIS4



Iceberg Catalog

ASF



Unity Catalog

LF



Hive Metastore

ASF



Schema Registry

CONFLUENT

LINEAGE1



OpenLineage

LF

OPERATIONS — HOW DATA IS QUERIED, OBSERVED, GOVERNED

QUERY2



SQL

ISO/IEC



SPARQL

W3C

DATA QUALITY3



dbt

DBT LABS



Great Expectations

GX LABS



SodaCL

SODA

OBSERVABILITY2



Open Telemetry

CNCF



OORS

BITOL @ LF

POLICIES1



OPA

CNCF

AI INTERFACES2



MCP

LF



A2A

LF

Cite this landscape